

Federated Learning for Ancestry-Aware Cancer Subtype Classification

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Abstract: Large-scale genomic datasets are biased towards individuals of European (EUR) ancestry, causing predictive models to generalise poorly across populations. We apply a federated learning (FL) framework to The Cancer Genome Atlas (TCGA) RNA-seq data for cancer subtype classification, treating two cohorts with different ancestry compositions as federated clients. Ancestry heterogeneity induces non-independent and non-identically distributed (non-IID) structure analogous to cross-silo FL, motivating federated optimisation as a principled solution. We compare five models — centralised ElasticNet, Federated Averaging (FedAvg), Federated Proximal (FedProx), ancestry-reweighted centralised, and their combination — evaluating overall predictive performance and ancestry-specific fairness across African (AFR), East Asian (EAS), South Asian (SAS), and admixed American (AMR) populations.

Keywords: Federated learning; Genomics; Ancestry; Non-IID data; Fairness.

1 Background and Methods

Large-scale genomic resources are strongly biased towards individuals of EUR ancestry, while AFR, EAS, SAS and AMR populations remain under-represented (Bick et al., 2024; Sohail et al., 2023). Models trained on such imbalanced data may learn ancestry-specific patterns rather than disease-relevant signals, perpetuating health disparities (Kolobkov et al., 2024). A key insight is that ancestry heterogeneity in genomic data closely parallels *cross-silo heterogeneity* in FL: local datasets follow different distributions and local model updates may diverge during training, a phenomenon known as client drift (McMahan et al., 2017; Li et al., 2020).

We use RNA-seq data from TCGA, accessed via cBioPortal (Gao et al., 2013), comprising 4,606 samples across ten cancer types (BRCA, HNSC,

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LGG, PRAD, THCA, LUAD, BLCA, LUSC, LIHC, STAD), with ancestry labels derived from admixture analysis (Carrot-Zhang et al., 2020), where global genetic ancestry is inferred from TCGA germline data and encoded as EUR, AFR, EAS, SAS, and AMR, used solely as analytical groupings for performance and fairness evaluation rather than as discrete biological categories. The ancestry distribution is EUR 3,674, AFR 447, EAS 419, AMR 35, SAS 31. After standardisation, we retain the 5,000 most variable genes as features; ElasticNet regularisation then performs the final sparse gene selection, selecting 1,895 genes in the centralised model. Two federated clients simulate realistic cross-silo heterogeneity: client C_1 receives 70% of all EUR samples plus 30% of non-EUR samples, resulting in a composition of 90% EUR, 5% AFR, 4% EAS, and 1% other; client C_2 comprises the remaining samples (63% EUR, 18% AFR, 17% EAS, and 2% other). The FL optimisation objective follows FedAvg (McMahan et al., 2017):

$$\min_w \sum_{k=1}^K \frac{n_k}{n} f_k(w), \quad n = \sum_{k=1}^K n_k,$$

where $f_k(w)$ is the local ElasticNet loss on client k . FedProx augments each local objective with a proximal term $(\mu/2)\|w - w_t\|^2$ ($\mu = 0.1$) to limit client drift (Li et al., 2020). Ancestry reweighting assigns each sample a weight inversely proportional to its ancestry group frequency, so underrepresented groups contribute equally to the loss. Table 1 summarises the five models compared. All models are evaluated on a held-out test set (20% stratified split) using macro-averaged one-vs-rest AUROC, balanced accuracy, and macro F1, reported both overall and per ancestry group. The fairness gap is defined as $\max_a \text{AUC}(a) - \min_a \text{AUC}(a)$ across ancestry groups a .

TABLE 1. Overview of models compared in this study.

Model	Name	Description
A	ElasticNet centralised	Pooled baseline
B	FedAvg	FL without proximal term
C	FedProx	FL with proximal term ($\mu = 0.1$)
D	Weighted centralised	Ancestry reweighting, no FL
E	Weighted + FedProx	Reweighting and proximal term combined

2 Results and Discussion

Table 2 reports overall and per-ancestry AUROC for all five models. Model A (centralised ElasticNet) achieves the highest overall AUROC (0.998) with a negligible fairness gap (0.000), reflecting strong generalisation of EUR-majority-trained features to cancer subtype classification. Model D (weighted

centralised) closely matches this performance (AUROC 0.997) while achieving perfect AFR AUROC (1.000) and the second-smallest fairness gap (0.001), making it the most practically useful model under a centralised setting. FL models B (FedAvg) and C (FedProx) incur an approximately 1% AUROC cost relative to centralised training (0.987 vs. 0.998), without a corresponding fairness benefit. Figure 1 shows that both FL variants exhibit increasing training loss after round 5, indicating client drift under the non-IID ancestry split; FedProx starts lower (0.15 vs. 0.16 at round 1) but converges to a similar level as FedAvg. Model E (Weighted + FedProx) shows substantially higher and unstable training loss throughout all 30 rounds, explaining its lower overall AUROC (0.976) and confirming that combining ancestry reweighting with federated training introduces conflicting gradient signals under this non-IID structure.

TABLE 2. AUROC (macro-averaged one-vs-rest) overall and per ancestry group. Best value per column in **bold**. [†]AUC not defined for AMR ($n=7$), EAS ($n=78$), SAS ($n=7$) due to insufficient class representation in the test set.

Model	Overall	EUR	AFR	BalAcc	Gap
A: ElasticNet centralised	0.998	1.000	1.000	0.990	0.000
B: FedAvg	0.987	0.987	0.980	0.930	0.007
C: FedProx	0.987	0.988	0.979	0.937	0.009
D: Weighted centralised	0.997	0.999	1.000	0.985	0.001
E: Weighted + FedProx	0.976	0.975	0.979	0.912	0.004

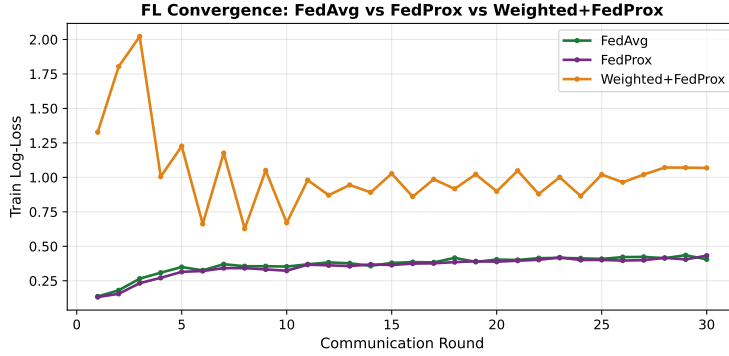


FIGURE 1. Training log-loss per communication round for FedAvg (B), FedProx (C), and Weighted+FedProx (E). Both B and C exhibit increasing loss after round 5, indicating client drift under the non-IID ancestry split. Model E shows substantially higher and unstable loss throughout, reflecting conflicting gradient signals from ancestry reweighting combined with federated optimisation.

These results indicate that ancestry reweighting without FL (Model D) provides the best trade-off between overall performance and minority-group fairness. The performance difference between FL and centralised models is negligible (AUROC gap of 0.011): FL achieves comparable predictive performance to centralised training while preserving data locality and avoiding the privacy risks of pooling raw genomic data across institutions (Kairouz et al., 2021). The similar performance across all models likely reflects the strong transcriptomic signal of cancer subtype classification (all models AUROC > 0.975), leaving little room for differentiation; weaker-signal tasks such as survival prediction may better reveal FL’s advantages. This conclusion is reinforced by follow-up experiments on BRCA with PAM50 subtypes, using one ancestry per federated client. A model trained on EUR samples alone generalises to AFR and EAS test sets with AUROC near 1.000, indicating that the transcriptomic subtype signal transfers across ancestries irrespective of the federated optimisation strategy. Future work will explore weaker-signal tasks, more extreme ancestry compositions, and deeper architectures.

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